

PRODUCT REQUIREMENTS DOCUMENT

Namma-Platform

Kannada-first Railway Guide for Rural Karnataka

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Platform	Android (Kotlin + Jetpack Compose)
Program	MindMatrix VTU Internship

1. Executive Summary

Namma-Platform is a Kannada-first Android app designed to solve a real problem faced by millions of rural railway passengers across Karnataka — confusion about which platform a train will arrive on, which coach to stand near, and whether the train is running on time.

The app targets passengers at small, unmanned, or semi-staffed railway stations where announcements are either inaudible, in English/Hindi only, or simply absent. It delivers the next 3 upcoming trains for any selected station, the platform number, a visual coach layout strip, a live train tracking map, and Kannada audio announcements — all for free.

Problem: Rural passengers at small stations cannot find platform/coach info in Kannada

Solution: Simple Kannada-first app: station → platform → coach layout → live map → audio

Stack: Kotlin · Jetpack Compose · OSMDroid · eRail API · Firebase · Android TTS

Cost: ₹0 to build and run (₹1,750 one-time only if published to Play Store)

Timeline: 6–8 weeks MVP · 3–6 months full feature parity

2. Problem Statement

2.1 The Real User Pain

At small railway stations in Karnataka (Tumkur, Arsikere, Hassan, Birur, Tiptur, etc.):

- Announcements are in Hindi or English — rural passengers, especially the elderly, cannot understand them
- Platform allocation changes last-minute, and there is no digital display board
- Passengers don't know where the General or Ladies coach will stop, so they crowd the wrong spot
- There is no way to tell if the train is 30 minutes late without asking a ticketing staff member
- Visually impaired and elderly passengers have no audio assistance in Kannada

2.2 Why Existing Apps Don't Solve This

App	What it does	Gap for rural user
NTES (Indian Railways)	Shows train running status	English only, complex UI, no coach layout
Where is my Train	Live tracking, delay alerts	English/Hindi, no Kannada TTS, no coach strip
RailYatri	Booking, tracking, food	Heavy app, requires login, not offline-capable

App	What it does	Gap for rural user
Namma-Platform (ours)	Kannada-first, platform + coach + map + TTS	Solves all gaps above — this is the product

3. Goals & Success Metrics

3.1 Product Goals

- G1 — Language Inclusion: Every key action must be performable in Kannada
- G2 — Zero Learning Curve: A 65-year-old first-time smartphone user can use the app in under 60 seconds
- G3 — Offline-First: Core features (station list, timetable) work with no internet
- G4 — Visual Clarity: High-contrast blue/yellow UI passes WCAG AA for low-vision users
- G5 — Live Awareness: User can see where the train currently is on a map

3.2 Success Metrics (MVP — 8 weeks post launch)

Metric	Target	Measurement
Task completion (find platform)	≥ 90% in < 30s	Usability test, 5 users
Kannada TTS plays correctly	100% of tested trains	Manual QA on 20 trains
Coach layout renders correct	All 5 tested trains	Visual comparison with IRCTC
Map loads station marker	< 3s on 4G	Firebase performance
Offline timetable works	No crash, stale data shown	Airplane mode test
App crash rate	< 1%	Firebase Crashlytics

4. Target Users & Personas

4.1 Primary Persona — Rukmini Amma

Age: 68 · Location: Arsikere, Hassan District · Occupation: Retired teacher

Device: Android Go (2GB RAM) · Data: BSNL prepaid, often spotty

Goal: Travel alone from Arsikere to Mysuru to visit daughter

Pain: Doesn't know which platform, misses train because she stood at wrong end of platform

Quote: "Announcement aagutide, aadre naige arthaagalla" (There's an announcement, but I don't understand it)

4.2 Secondary Persona — Ravi (Commuter)

Age: 27 · Location: Tumkur · Occupation: Factory worker, daily commuter

Device: Mid-range Android · Data: Jio 4G

Goal: Know if the 7:15 AM Bangalore express is late without asking anyone

Pain: NTES is confusing, Where Is My Train drains battery, no Kannada support

Quote: "App simple aagirali, nanna amma kuda use maadali" (App should be simple, even my mom should use it)

5. Feature List (A–Z Scope)

5.1 Feature Priority Matrix

Feature	Priority	Phase	Effort
Station Selection	P0 — Must Have	MVP	Low
Next 3 Trains Display	P0 — Must Have	MVP	Low
Platform Number Display	P0 — Must Have	MVP	Low
Coach Layout Strip	P0 — Must Have	MVP	Medium
Kannada TTS Announcement	P0 — Must Have	MVP	Low
Live Train Map Tracker	P1 — Should Have	MVP	High
Train Delay Alert	P1 — Should Have	MVP	Medium
Offline Mode (local JSON)	P1 — Should Have	MVP	Medium
Kannada / English Toggle	P1 — Should Have	MVP	Low
Train Search by Number	P2 — Nice to Have	Phase 2	Low
PNR Status Lookup	P2 — Nice to Have	Phase 2	Medium
Push Notifications (delays)	P2 — Nice to Have	Phase 2	Medium
Seat Availability	P3 — Future	Phase 3	High

6. Detailed Feature Specifications

Feature 1 — Station Selection Screen

The entry point of the app. User selects their current station from a searchable list.

User Story

As Rukmini Amma, I want to select my station by typing 2-3 Kannada or English letters, so I can quickly find Arsikere without scrolling.

Acceptance Criteria

- Search bar auto-focuses on screen open; keyboard visible immediately
- Supports both Kannada script (ಅರಸೀಕೆರೆ) and English transliteration (Arsikere)
- Results filter live as user types with debounce of 300ms
- Station list pre-loaded from local JSON — works fully offline
- Recently selected stations shown at top (max 3, stored in DataStore)
- Empty state shows friendly Kannada message if no match found

UI Spec

- Top app bar: 'ನಿಮ್ಮ ನಿಲ್ದಾಣ ಆರಿಸಿ / Select Your Station'
- Search field: Material3 OutlinedTextField, yellow background (#FFF176), blue border
- Station list: LazyColumn, each item shows station name (Kannada large) + code (English small)

Feature 2 — Next 3 Trains Dashboard

The core screen. After selecting a station, shows the next 3 trains arriving/departing, with platform numbers.

User Story

As Ravi, I want to see the next 3 trains from Tumkur to Bangalore with their platform numbers on one screen, so I know where to stand.

Acceptance Criteria

- Shows exactly 3 upcoming trains (or fewer if schedule ends for the day)
- Each card shows: Train name (Kannada + English), Train number, Departure time, Platform number, Delay status
- Platform number shown in large bold font (min 40sp) with high-contrast yellow chip
- Data fetched from eRail API; refreshes every 5 minutes automatically
- Pull-to-refresh supported
- If API fails, shows cached data with 'Offline — showing last known data' banner
- Shimmer loading skeleton shown while fetching

UI Spec

- Card: Material3 ElevatedCard, white background, blue left border accent (4dp)
- Platform chip: Yellow (#FDD835) background, blue (#1565C0) bold text, rounded 8dp
- Delay indicator: Green dot (on time) / Red dot (delayed) with minute count

Feature 3 — Coach Layout Strip

A horizontal scrollable visual strip showing where each coach type stops on the platform.

User Story

As Rukmini Amma, I want to see a picture of the train showing where the 'General' coach is, so I walk to the right spot on the platform without guessing.

Acceptance Criteria

- Horizontal scrollable strip using LazyRow
- Each coach shown as a rectangle with label: Engine, General (S), Ladies, S1, S2, B1, etc.
- General and Ladies coaches highlighted in bright yellow; other coaches in blue
- Strip scrollable from Engine (left) to last coach (right)
- Coach data stored in local JSON keyed by train number
- 'Stand Here' indicator arrow overlaid on General/Ladies section
- Tapping any coach shows a tooltip: what kind of passengers board here

Data Model (local JSON)

```
{
  "trainNumber": "16557",
  "trainName": "Rajya Rani Express",
  "coaches": ["Engine","GEN","GEN","S1","S2","S3","S4","S5","B1","B2","A1"]
}
```

Feature 4 — Live Train Map Tracker

A map view showing where the selected train currently is, with an animated train icon moving station-to-station.

User Story

As Ravi, I want to see my train's position on a map like Google Maps shows my Uber, so I know when to start walking to the platform.

How It Works (Technical)

```
Step 1 → App calls eRail API: /live-train-status/{trainNo}/{date}
Step 2 → Response includes: lastStation, nextStation, delayMinutes, updatedAt
Step 3 → App places animated marker at lastStation coordinates on OSMDroid map
```

Step 4 → Marker smoothly animates toward nextStation using ValueAnimator over 5 minutes

Step 5 → API is polled again every 5 minutes; marker repositions to new lastStation

Result → User sees a train icon 'moving' along the route — identical feel to live GPS

Acceptance Criteria

- Map rendered using OSMDroid (OpenStreetMap) — zero API cost
- Railway track layer loaded from OpenRailwayMap tile server — free
- Train marker is a custom Compose-drawn icon (train emoji or vector drawable)
- Marker animates smoothly between stations using ObjectAnimator on LatLng
- Map centers on train marker automatically; user can pan freely
- Shows list of upcoming stations in a bottom sheet: station name + ETA
- 'Last updated X minutes ago' shown; auto-refreshes every 5 minutes
- Works on 2G — tiles cached by OSMDroid; minimal data usage

Map UI Spec

- OSMDroid MapView embedded in Compose via AndroidView
- OpenFreeMap tiles as base layer (no API key needed)
- OpenRailwayMap overlay layer on top for rail track visualization
- Bottom sheet (BottomSheetScaffold): upcoming stations list
- FAB: 'Re-center on train' button

Feature 5 — Kannada TTS Announcements

A 'Help Me' button that speaks the next train's platform and coach information aloud in Kannada.

User Story

As Rukmini Amma, I want the app to tell me in Kannada which platform my train is on, so I don't have to read anything.

Acceptance Criteria

- Large 'Help Me / ಸಹಾಯ ಮಾಡಿ' button visible on the train dashboard screen
- On tap: TTS reads out: 'Bangalore Express, platform eradu, general coach train mundhe iruttade'
- Uses Android TextToSpeech API with Locale('kn', 'IN') — zero cost, works offline
- If Kannada TTS not installed: shows dialog with link to download Kannada TTS pack
- Volume at max by default; respects system volume
- Repeat button: plays announcement again without re-tapping Help Me

TTS Script Template

Template: "{trainName}, platform {platformNo}, {coachType} dabba nimma {position} iru"

Example: "Bangalore Express, platform eradu, general dabba nimma mundhe iru"

English: "Bangalore Express, platform 2, general coach is ahead of you"

Feature 6 — Offline Mode

The app continues to show timetable and coach data even with no internet connection.

Acceptance Criteria

- Station list always loaded from local JSON bundled in assets/
- Static timetable for top 50 Karnataka stations bundled as fallback data
- When online: data synced to Room database (local SQLite)
- When offline: Room database serves cached data
- Banner shown: 'ಇಂಟರ್ನೆಟ್ ಇಲ್ಲ — ಹಳೆಯ ಮಾಹಿತಿ ತೋರಿಸಲಾಗುತ್ತಿದೆ' (No internet — showing last known data)
- Map tracker gracefully degrades: shows last known position, stops animating

7. Complete Technology Stack

7.1 Stack Overview

Layer	Technology	Why Chosen	Cost
Language	Kotlin	Official Android language, coroutines	Free
UI Framework	Jetpack Compose	Modern declarative UI, less boilerplate	Free
Architecture	MVVM + Clean Arch	Separation of concerns, testable	Free
Maps	OSMDroid	OpenStreetMap for Android, no API key	Free
Map Tiles	OpenFreeMap	Beautiful tiles, unlimited, no key	Free
Rail Overlay	OpenRailwayMap	Shows actual rail tracks	Free (small apps)
Train Data	eRail.in API	Free key, 7500+ trains, live status	Free
Backend DB	Firebase Realtime DB	Spark plan: 1GB free, real-time sync	Free
Auth	Firebase Auth	Anonymous auth for Spark free tier	Free

Layer	Technology	Why Chosen	Cost
Local DB	Room (SQLite)	Offline cache, typed queries	Free
Preferences	DataStore (Proto)	Replaces SharedPreferences, async	Free
Networking	Retrofit + OkHttp	Type-safe HTTP, interceptors	Free
JSON	Kotlin Serialization	Fast, Kotlin-native JSON parsing	Free
Images/Icons	Coil	Compose-native image loading	Free
DI	Hilt	Jetpack-official dependency injection	Free
TTS	Android TextToSpeech	Built-in, Kannada locale support	Free
Design System	Material 3	Google's latest Compose components	Free
Analytics	Firebase Analytics	Spark free tier, crash reports	Free
Crash Reports	Firebase Crashlytics	Free, real-time crash logs	Free
IDE	Android Studio	Official, free	Free
Version Control	GitHub	Free public repo	Free

7.2 Architecture Diagram (MVVM + Clean Architecture)

```

Presentation Layer — Jetpack Compose Screens → ViewModels → StateFlow/UiState
|
Domain Layer ————— UseCases (GetNextTrainsUseCase, GetTrainPositionUseCase, etc.)
|
Data Layer ————— Repositories → [Remote: Retrofit + eRail API] + [Local: Room DB]
|
Infrastructure ———— Firebase (sync) · OSMDroid (map) · Android TTS · DataStore

```

7.3 Key Dependencies (build.gradle)

```
// Compose BOM
```

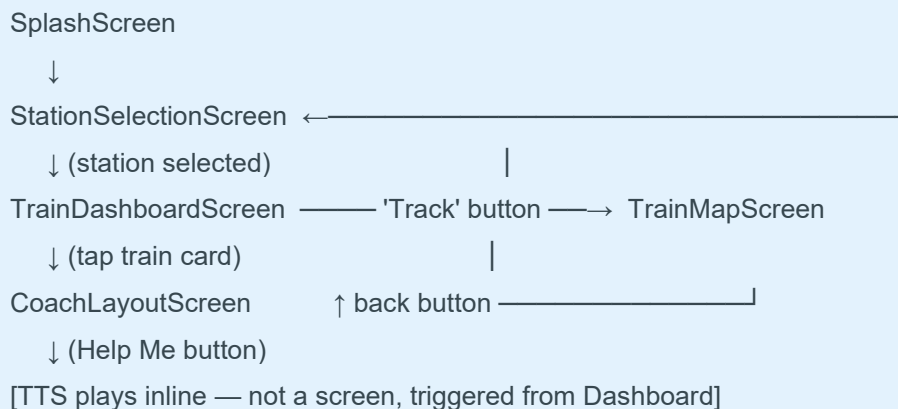
```

implementation(platform("androidx.compose.compose-bom:2024.06.00"))
implementation("androidx.compose.ui:ui")
implementation("androidx.compose.material3:material3")
// Maps
implementation("org.osmdroid:osmdroid-android:6.1.18")
// Networking
implementation("com.squareup.retrofit2:retrofit:2.11.0")
implementation("com.squareup.retrofit2:converter-kotlinx-serialization:2.11.0")
// Firebase
implementation(platform("com.google.firebase:firebase-bom:33.1.0"))
implementation("com.google.firebase:firebase-database-ktx")
// Room
implementation("androidx.room:room-runtime:2.6.1")
ksp("androidx.room:room-compiler:2.6.1")
// Hilt
implementation("com.google.dagger:hilt-android:2.51.1")
ksp("com.google.dagger:hilt-android-compiler:2.51.1")
// DataStore
implementation("androidx.datastore:datastore-preferences:1.1.1")

```

8. Screen Flow & Navigation

8.1 Navigation Graph



8.2 Screen Inventory

Screen	Key Components	Route Name
Splash Screen	App logo, version, auto-navigate after 1.5s	splash
Station Selection	SearchBar, LazyColumn, Recent stations	station_selection
Train Dashboard	TrainCard x3, PlatformChip, HelpMeButton, RefreshIndicator	train_dashboard/{stationCode}
Coach Layout	LazyRow coach strip, StandHereIndicator, TooltipPopup	coach_layout/{trainNo}
Train Map	OSMDroid MapView, TrainMarker, BottomSheet, FAB	train_map/{trainNo}/{date}

9. API Contracts & Data Models

9.1 eRail API Endpoints Used

Endpoint	Purpose	Response Key Fields
GET /trains-between-stations/	Fetch next trains for a station	trainNo, trainName, departureTime, platformNo
GET /live-train-status/	Live position + delay of a train	currentStation, nextStation, delayMins, updatedAt
GET /train-schedule/	Full route + station list of a train	stations[]: {stationCode, arrivalTime, distKm}

9.2 Key Data Classes (Kotlin)

```
data class Train(
    val trainNo: String,
    val trainName: String,
    val departureTime: String,
    val platformNo: Int,
    val delayMinutes: Int,
    val coaches: List<Coach>
)
```

```
data class TrainPosition(
    val trainNo: String,
```

```

val lastStation: Station,
val nextStation: Station,
val delayMinutes: Int,
val updatedAt: Long
)

data class Station(
    val code: String,
    val nameKannada: String,
    val nameEnglish: String,
    val lat: Double,
    val lng: Double
)

```

10. UI Design System

10.1 Color Palette (High Contrast — WCAG AA Compliant)

Token	Hex	Usage	Contrast
Primary Blue	#1565C0	Buttons, headers, accents	AAA on white
Platform Yellow	#FDD835	Platform chips, coach highlight	AA on dark text
Background	#FAFAFA	Screen background	—
Surface	#FFFFFF	Cards, bottom sheets	—
On Primary	#FFFFFF	Text on blue buttons	AAA
Error Red	#B71C1C	Delay indicator, errors	AA on white
Success Green	#2E7D32	On-time indicator	AA on white
Text Primary	#212121	Body text, labels	AAA on white
Text Secondary	#616161	Subtitles, hints	AA on white

10.2 Typography

Style	Font	Size	Usage
displayLarge	Noto Sans Kannada	57sp	Platform number

Style	Font	Size	Usage
headlineMedium	Noto Sans Kannada	28sp	Train name (Kannada)
titleMedium	Roboto	16sp	Train name (English)
bodyMedium	Noto Sans Kannada	14sp	Station names, labels
labelSmall	Roboto Mono	11sp	Train number code

10.3 Component Library

- TrainCard — ElevatedCard with blue left border, platform chip, delay dot
- PlatformChip — Yellow rounded box, big bold number, min touch 48dp
- CoachBox — Colored rectangle in LazyRow; yellow for General/Ladies, blue for others
- HelpMeButton — Large FAB, mic icon, Kannada label, fires TTS
- DelayBadge — Green/Red dot + minutes text, inline in train card
- TrainMarker — Custom Compose Canvas drawn icon for OSMDroid overlay
- OfflineBanner — Amber banner at top of screen when no network detected

11. Non-Functional Requirements

11.1 Performance

- Cold start time: < 2 seconds on Android Go (2GB RAM, Android 11)
- API response → UI render: < 3 seconds on 4G, < 8 seconds on 2G
- Map tile load: OSMDroid tile caching enabled; tiles cached for 7 days
- TTS first-word latency: < 500ms
- Memory usage: < 120MB RAM during map view

11.2 Compatibility

- Minimum SDK: API 24 (Android 7.0) — covers 94% of Indian Android devices
- Target SDK: API 35 (Android 15)
- Screen sizes: 4.5" to 6.7", portrait only (landscape lock for MVP)
- Works on Android Go Edition (low-RAM mode)

11.3 Security & Privacy

- No personal data collected — Firebase Anonymous Auth only
- No location permission required — map shows train position, not user position
- API key stored in local.properties, not hardcoded or committed to Git

- Network calls over HTTPS only

11.4 Accessibility

- All interactive elements have contentDescription in Kannada and English
- Minimum touch target: 48dp × 48dp (Material 3 standard)
- TTS is the primary accessibility feature — built for low-literacy users
- High contrast mode supported via Material 3 dynamic color

12. Development Roadmap

Phase 1 — MVP (Weeks 1–6)

Week	Tasks	Deliverable
Week 1	Project setup · Android Studio · Hilt · Room · Retrofit · Navigation Compose · Firebase init · eRail API key registration	Project skeleton builds
Week 2	Station JSON · StationSelectionScreen · Search + filter · DataStore recent stations	Station screen works
Week 3	eRail API integration · Repository layer · TrainDashboardScreen · Train cards · Platform chips · Delay badges	Live trains display
Week 4	Coach JSON data · CoachLayoutScreen · LazyRow strip · Highlight logic · StandHere indicator · Tooltips	Coach strip works
Week 5	OSMDroid integration · OpenFreeMap + OpenRailwayMap tiles · Train marker · ValueAnimator station-to-station · Bottom sheet upcoming stations	Map tracker works
Week 6	Kannada TTS · TextToSpeech engine · Script templates · HelpMeButton · Offline Room cache · OfflineBanner · Polish + bugfix + testing	MVP APK ready

Phase 2 — Scale (Months 2–4)

- Expand station database from 50 → 500+ Karnataka stations
- Add train search by number
- Firebase Cloud Messaging for delay push notifications
- Hindi and Telugu TTS support
- PNR status lookup screen

Phase 3 — Platform (Months 5–12)

- All-India station coverage
- GenAI coach suggestion ("Stand near coach S3 for shorter exit walk at your destination")
- Crowd-sourced platform reports ("Platform 2 is crowded right now")
- Play Store publish (one-time ₹1,750 fee)

13. Risks & Mitigations

Risk	Impact	Mitigation
eRail API goes down or changes	App shows no live data	Fallback to static timetable JSON bundled in app
Kannada TTS not installed on device	No audio announcements	Show dialog to install Kannada TTS pack from Play Store
OSMDroid tiles slow on 2G	Map doesn't load fast enough	Pre-cache tiles for top 20 stations at app install
IRCTC coach data changes	Wrong coach layout shown	Coach JSON versioned; in-app update via Firebase Remote Config
eRail API rate limit hit	API calls blocked	Implement request caching; 5-min refresh interval (not real-time)

14. Open Questions

- Q1: Should the app support landscape mode for elderly users with larger phones? (Defer to Phase 2)
- Q2: Should coach position data be crowdsourced or manually curated? (Manual for MVP, crowdsource later)
- Q3: Do we need a backend proxy for eRail API key security? (No for MVP — key in local.properties, never in repo)
- Q4: Is Noto Sans Kannada font bundled in app or relying on system font? (Bundle it — Android Go may not have it)

15. Glossary

Term	Definition
NTES	National Train Enquiry System — official Indian Railways live status website
eRail API	Third-party Indian Railways API at api.erail.in — free tier available
OSMDroid	OpenStreetMap SDK for Android — maps without Google Maps API key
Coach Strip	Visual horizontal layout of train coaches shown in order from engine to end
Platform Chip	Yellow badge showing platform number, prominently sized for visibility
TTS	Text-to-Speech — Android API that reads text aloud using Kannada voice
DataStore	Jetpack library for storing key-value preferences, replaces SharedPreferences
Room	Jetpack SQLite ORM — used for offline caching of train and station data
Hilt	Google's dependency injection framework for Android, built on Dagger
MVVM	Model-View-ViewModel — architecture pattern used throughout the app
Station-hop tracking	Simulated live tracking: train marker moves between last-known stations, updated every 5 min via API

End of PRD — Namma-Platform v1.0.0
MindMatrix VTU Internship Program · Bengaluru, Karnataka